

Minia University

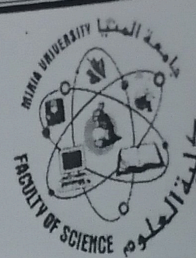
Department of Mathematics

Course : أساسيات التحليل العددي

Faculty of Science

Final 2nd Term Exam 2023/2024

Code: M208



(Total: 70 Marks)

Answer the following questions:

Question 1: (30 Marks)

A) Using two different methods, find the real root of the nonlinear equation:

$$x^3 = 2x + 5, \quad x_0 = 2.5 \quad (\text{The accuracy of the root is 3 digits})$$

B) From the following table:

	x_0	x_1	x_2
x	-1	2	5
$f(x)$	6	3	9

Find: (i) $f'(1)$, (ii) $\int_{-1}^5 f(x) dx$.

Question 2: (25 Marks)

A) From the following table:

x	0.0	0.2	0.4	0.6	0.8	1.0
$y = f(x)$	4	4.08	4.64	6.16	9.12	14

Find: (i) $f(0.3)$, (ii) $f'(0.8)$.

b- Evaluate $\int_4^{5.2} \sqrt{x} dx$, by using the Simpson's $\frac{1^{rd}}{3}$ rule, the Simpson's $\frac{3^{th}}{8}$ rule and find the error.

x	4	4.6	5.2
y			

Question 3: (15 Marks)

Solve the following system of simultaneous linear equations by using the Gauss seidel iteration method:

$$2x - 3y + 20z = 25,$$

$$20x + y - 2z = 17,$$

$$3x + 20y - z = -18.$$

Best wishes and wish Good Luck

Dr.: Hela Hamdy Mohamed

Gauss
Jacobi